THE UNIVERSITY OF HONG KONG

DEPARTMENT OF MATHEMATICS

MATH2012: Fundamental Concepts of Mathematics

Dec 21, 2022 2:30 p.m. – 5:00 p.m.

Only approved calculators as announced by the Examinations Secretary can be used in this examination. It is candidates' responsibility to ensure that their calculator operates satisfactorily, and candidates must record the name and type of the calculator used on the front page of the examination script.

Answer ALL SIX questions

Note: You should always give precise and adequate explanations to support your conclusions. Clarity of presentation of your argument counts. So **think carefully before you write.**

1. (30 points) Prove or disprove each of the following statements:

   (a) There exist $a, b, c, d \in \mathbb{N}$ such that $ab = cd$ and $a + b + c + d$ is prime.

   (b) For any $a, b \in \mathbb{Q}$ with $a \neq b$, if $\sqrt[3]{a} + \sqrt[3]{b} \in \mathbb{Q}$, then $\sqrt[3]{a} \in \mathbb{Q}$ and $\sqrt[3]{b} \in \mathbb{Q}$.

   (c) Let R be a relation on $\mathbb{N}$ defined by

   $$xRy \text{ if } \frac{2x^2}{y} \in \mathbb{N} \text{ or } \frac{2y^2}{x} \in \mathbb{N}.$$

   Then R is an equivalence relation on $\mathbb{N}$.

   (d) For any nonempty sets A, B and for any function $f : A \longrightarrow B$, if f has a left inverse $g : B \longrightarrow A$ and a right inverse $h : B \longrightarrow A$, then $g = h$.

   (e) There exists a nonempty countable set A such that there is no surjective function $f : \mathbb{N} \longrightarrow A$.

2 (12 points)

   (a) Prove that for any $n \in \mathbb{N}$, $5^n \equiv 1 \pmod 8$ if and only if n is even.

   (b) Hence, or otherwise, find all $n \in \mathbb{N}$ such that $4^n + 24^n + 501^n$ is a perfect square.

3. (10 points) Use induction to prove that for any integer $n \geq 6$, there exist $x_1, x_2, \ldots, x_n \in \mathbb{N}$ such that the sum of the reciprocals of their squares is equal to 1, that is,

$$\frac{1}{x_1^2} + \frac{1}{x_2^2} + \cdots + \frac{1}{x_n^2} = 1.$$

4. (16 points) Let $f : \mathbb{R} \longrightarrow \mathbb{R}$ be a function satisfying

$$f(xf(x) + f(y)) = (f(x))^2 + y \quad \text{for any } x, y \in \mathbb{R}. \qquad (*)$$

   (a) Prove that f is surjective. Hence, or otherwise, prove that $f \circ f = id_{\mathbb{R}}$ and $f(0) = 0$.

   (b) Find all functions $f : \mathbb{R} \longrightarrow \mathbb{R}$ satisfying $(*)$.

5. (16 points) Let A and B be nonempty sets.

(a) Let $f : A \longrightarrow B$ be a function. Define a function $F : \mathcal{P}(A) \longrightarrow \mathcal{P}(B)$ by $F(X) = f(X)$ for any $X \in \mathcal{P}(A)$. Prove that

(i) f is injective if and only if F is injective.

(ii) f is surjective if and only if F is surjective.

(b) Prove that if A is numerically equivalent to B, then $\mathcal{P}(A)$ is numerically equivalent to $\mathcal{P}(B)$.

6. (16 points) Let A and B be nonempty sets and let $f : A \longrightarrow B$ be a function.

(a) Prove that for any subsets $Y_1, Y_2 \subseteq B$,

$$f^{-1}(Y_1 \cap Y_2) = f^{-1}(Y_1) \cap f^{-1}(Y_2).$$

(b) Suppose that P is a partition of B. Let

$$Q = \{X \subseteq A : X \neq \emptyset \text{ and } X = f^{-1}(Y) \text{ for some } Y \in P\}.$$

Prove that for any $X_1, X_2 \in Q$, either $X_1 = X_2$ or $X_1 \cap X_2 = \emptyset$. Hence prove that Q is a partition of A.

(c) Let $A = B = \mathbb{Z}$ and let $f : \mathbb{Z} \longrightarrow \mathbb{Z}$ be defined by $f(x) = 6x^2+7$. Consider the partition $P = \mathbb{Z}_8 = \{[0]_8, \ldots, [7]_8\}$ of B. Use (b) to construct the corresponding partition Q on A.

******* **END OF PAPER** *******